



MODULE NAME:	MODULE CODE:
CLOUD DEVELOPMENT B	CLDV7112/w

ASSESSMENT TYPE: PROJECT 1 (PAPER AND MARKING RUBRIC)
TOTAL MARK ALLOCATION: 100 MARKS
TOTAL HOURS: A MINIMUM OF 15 HOURS IS SUGGESTED TO COMPLETE THIS ASSESSMENT

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

2. **No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.**
3. **Make a copy of your assignment before handing it in.**
4. *Assignments must be typed unless otherwise specified.*
5. *Begin each section on a new page.*
6. *Follow all instructions on the cover sheet.*
7. *This is an individual assignment.*
8. *Answer All Questions.*
9. *Instructions for submitting your assessment:*
 - a. *Use of good programming practice and comments in code is compulsory.*
 - b. *Save your solution/project files in your GitHub repository for this module.*
 - c. *Save all files (including any source code files, template files, design files, image files, text files, database files, etc.) within your GitHub repository.*
 - d. *Do NOT save zipped (archive) files in your GitHub repository unless specifically instructed to do so.*
 - e. **Important:** *Upon completion of your assessment, you must save and close all your open files before submitting your work. You will submit your assessment on the LMS page for this module.*
 - f. **To complete your submission:** *Create a document in MS-Word or Notepad. The document name must follow the format explained in the instructions.*
 - g. *In this document, include the following: Your student number, the module code, all required answers, **the link to your GitHub repository** where you saved your practical work, and the URL of your web application.*
 - h. *Submit this document in the LMS, using the submission links for this module.*

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where both minor and major errors** (e.g., two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from the overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g., alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g., publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Background

ABC Retail, a rapidly growing online retailer, currently manages its order processing system using an aging **on-premises** infrastructure. They store customer orders and product information in a **traditional relational database** system, which struggles to handle the increasing volume of transactions during peak shopping seasons like Christmas and other related holidays.

Additionally, the company stores product images in **network shared drives**, leading to storage inefficiencies and slow access times. The message queuing system, powered by **legacy middleware**, lacks scalability and reliability, often resulting in message delivery delays and processing errors. As a result, ABC Retail is facing customer complaints, missed sales opportunities, and operational inefficiencies.

Despite migrating their order processing system to the cloud, ABC Retail continues to face challenges in handling **real-time event processing** and reliable **message queuing**. Their current setup relies on a combination of custom-built event processing pipelines and third-party messaging solutions, which lack the scalability and flexibility needed to support the company's growing business demands. As a result, ABC Retail experiences delays in order processing, inconsistent messaging delivery, and difficulty in maintaining and scaling their existing infrastructure.

ABC Retail's current **data analytics** infrastructure struggles to keep pace with the growing volume and complexity of customer data. **Traditional relational databases** and **on-premises analytics tools** are unable to efficiently process and analyse diverse data types, leading to delays in generating actionable insights. As a result, ABC Retail faces challenges in personalizing customer experiences, optimizing product recommendations, and improving operational efficiency.

To work on this Project, students are required to:

- Have access to an Azure account with available credit. This access will be arranged by your lecturer/campus at the start of this module.
- Use Microsoft Visual Studio for your coding.
- Save source code in a GitHub repository. This access will be arranged by your lecturer/campus at the start of this module.

The submission for this Project will require you to do the following:

- Create a **document** in MS-Word which contains the following:
 - Your student number.
 - The module code.
 - All answers required, including typed answers, diagrams, and screenshots.
 - The ***URL of the Web App*** that you developed.
- The document name must follow the format shown here:
 - ***StudentNumber_ModuleCode_Project#.***
 - E.g., if your student number is *12345* and you are submitting *Project 1* for the module *PROG121*, create a document named ***12345_Prog121_Project1.***
- Submit this document in Turnitin, using the submission link on the Turnitin page for this module.

Summary Sheet

Item	Description
Summary of Activities	The student needs to submit this Project on the Learn/LMS page for this module. <i>NB. Please follow the given instructions to supply a document that includes the needed answers, screenshots, the URL of the Web App module developed, and the link to the GitHub repository containing your project source code.</i>
Tools and Resources	• Microsoft Visual Studio. • Microsoft SQL Server. • Microsoft Word or other word processing software. • Windows Azure Portal. • Windows Azure subscription with Microsoft Azure Storage. • A GitHub repository.
Calculation of Marks	The final module mark will be weighted as follows: • Project 1 – 25% (formative) • Project 2 – 30% (formative) • ICE Tasks – 10% • Practicum – 35% (summative)

Project 1 — Azure Storage Solution**(Marks: 100)**

Related Content: Learning Units: 1 – 3

Assessment

Assessment/ Deliverable	Marks	Weight	Duration
Project 1	100	25%	15 hours

Your submission document

Your submission for this Project must be submitted in an MS-Word document, containing the following:

- Your student number.
- The module code.
- The URL of your deployed application.
- The GitHub link for the web application source code.
- Screenshots showing your successful implementation of the functionalities as described below, as well as screenshots of the deployed web application.

Note: Make sure you read the instructions given earlier in this document and follow the specified file name format.

Develop an application that makes use of various Azure Storage Services.

You are required to develop a web application with the following capabilities:

- Store customer profiles and product-related information using Azure Tables.
- Host images and multimedia content using Azure Blob Storage.
- Details relating to the processing of orders and inventory management should be stored in Azure Queues. An example would be uploading an image “imageName”, “Processing order”, etc.
- Log files should be stored using Azure Files.

In your Azure services, you are required keep in mind scalability, reliability, and cost-effectiveness. In your web applications, you are required to use the relevant controls to upload or download, and display the data stored in your Azure Storage Services.

Once you have tested your web application, you are required to deploy it to an Azure App Service and make it accessible through a link. Ensure that this has been tested both on your local computer and on the App Service.

Submitting your document for this Project

You are required to submit a single MS-Word document with the following:

- Screenshots of your various storage services, with at least 5 records each, as well as screenshots of the deployed web application
- The URL for your deployed Web Application
 - The URL should follow the following format and must look similar to this:
http://student_number.azurewebsites.net
- The GitHub link for the web application source code

Once you have completed your submission doc for this Project, submit your document using the submission link on the Turnitin page for this module.

Note: Make sure you read the instructions given earlier in this document and follow the specified file name format.

Rubric 1 (for Project 1)					Comments
	Excellent	Good	Developing	Poor	
Able to store information successfully into a Table Storage using the appropriate controls in a web application. Screenshots showing at least 5 records are provided.	16 – 20	11 – 15	6 – 10	0 - 5	
Multimedia information is successfully stored in a Blob Storage, using the appropriate web application controls. Screenshots showing at least 5 records are provided.	16 – 20	11 - 15	6 – 10	0 - 5	
Queues have been used to store transactions and the inventory process using the correct format. Screenshots showing at least 5 records are provided.	16 – 20	11 - 15	6 – 10	0 - 5	
Files have been stored in an Azure File service using their file names. Screenshots showing at least 5 records are provided.	16 – 20	11 - 15	6 – 10	0 - 5	
The web application is deployed to an Azure App Service and runs correctly in the online environment. - Screenshots of the Azure deployment process are provided. - URL is provided and accessible in a web browser.	16 – 20	11 - 15	6 – 10	0 - 5	
Project 1 Total					/100

[TOTAL MARKS: 100]